MIL-R-43870B
26 November 1986
SUPERSEDING
MIL-R-43870A(GL)
7 September 1983

MILITARY SPECIFICATION

RACK, STORAGE, MIXER BOWLS AND ATTACHMENTS, MOBILE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a mobile rack with two shelves and three cross members with attachment pegs suitable for hanging and storing mixer bowls and attachments (see 6.1).

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

FF-C-77	- Casters, Rigid and Swivel (Institutional Duty)
QQ-A-200/9	- Aluminum Alloy 6063, Bar, Rod, Shapes, Tube, Wire Extruded

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7320

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QQ-A-250/2 - Aluminum Alloy 3003, Plate and Sheet
QQ-A-250/8 - Aluminum Alloy 5052, Plate and Sheet
PPP-B-591 - Box, Fiberboard, Wood Cleated
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Boxes, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
A 269 - Seamless and Welded Austenitic Stainless Steel Tubing for General Service
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL SANITATION FOUNDATION (NSF)

Standard No. 2 - Food Service Equipment

Current Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, 3475 Plymouth Road, P.O. Box 1468, Ann Arbor, MI 48106.)

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENT

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Standard product. The racks delivered under this document shall be the manufacturer's standard commercial product, except for any changes necessary to comply with document requirements. All like items furnished on any one contract, including parts and subassemblies thereof, shall be new and interchangeable.

3.3 Codes and standards. The racks shall comply with the applicable requirements of NSF Standard No. 2 (see 4.5).

3.4 Materials and components. Materials and components not definitely specified shall be of the quality normally used by the manufacturer for his racks provided the completed item complies with all provisions of this document. Recycled material may be used (see 6.4).

3.4.1 Stainless steel and aluminum.

3.4.1.1 Sheet and strip. Stainless steel sheet, plate and strip shall conform to the requirements for types 302 and 304 of ASTM A 167. Aluminum sheet and plate shall conform to the requirements of QQ-A-250/8.

3.4.1.2 Tubing. Aluminum or stainless steel tubing shall conform to the requirements of temper 5 of QQ-A-200/9 and type 304 of ASTM A 269, respectively.

3.4.1.3 Channel. Aluminum channel shall conform to the requirements of temper 5 of QQ-A-200/9.

3.4.2 Threaded fasteners. Threaded fasteners shall be stainless steel or carbon steel plated with nickel, chromium, or zinc.

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* 3.5 Design and construction. The rack shall consist of a tubular framework with two shelves and three upper cross members with attachment pegs and four swivel casters with revolving bumpers, as shown in figure 1. Figure 1 is for illustrative purposes only and is not intended to preclude acquisition of racks otherwise meeting all the requirements of this document. The rack shall be capable of being taken apart, with shelves and the upper cross member assembly removable. Maximum overall dimensions shall be 55 inches in length, 30 inches in width, and 66 inches in height. The item shall show no evidence of racking in excess of 1/2 inch, parts looseness, separation, breakage, or permanent deformation when tested as specified in 4.6.1, 4.6.2, and 4.6.3.

* 3.5.1 Frame. Uprights and cross members shall be stainless steel tubing with a minimum of 1 1/4 inches in diameter by 0.056 inch wall or a combination of extruded aluminum tubing, 1 1/16 inch by 2 inches by 0.083 inch thick wall, for the uprights and extruded aluminum channel, 2 3/8 inches wide by 1 inch by 0.187 inch thick wall, for the cross members conforming to the material specified in 3.4.1.2 and 3.4.1.3. The rear upright shall consist of two sections: the upper section shall have three cross members welded across them, equally spaced 6 to 8 inches apart, forming the removable cross member assembly. The frame shall have cross-bracing, if necessary, to eliminate racking. Cross-bracing members shall be fabricated from stainless steel or aluminum.

* 3.5.2 Shelves. Two shelves measuring not less than 24 by 48 inches shall be provided. The top shelf shall be fabricated of stainless steel. The bottom shelf shall be stainless steel or aluminum. Shelves constructed of aluminum shall be a minimum of 0.187 inch thick and shall conform to QQ-A-250/2, temper H14 or QQ-A-250/8, temper H32. Shelves constructed of stainless steel shall be a minimum of 0.071 inch thick and shall conform to 3.4.1.1. The upper shelf shall be 33 to 36 inches above the floor and the lower shelf 8 to 11 inches. The shelves shall be of the slotted or louvered type design to allow for air drying. The shelves shall be attached to the uprights by means of threaded fasteners, and each shall be capable of supporting a static load of 120 pounds.

* 3.5.3 Attachment pegs. The upper and middle cross members shall have four and three attachment pegs, respectively, with offset positioning as shown in figure 1. These pegs shall be minimum 6-inch lengths of stainless steel tubing that is minimum 3/4-inch diameter by 0.056-inch thick wall or 1/2-inch schedule 40 aluminum pipe welded to the cross member, protruding upward approximately at a 45 degree angle. The lower cross member shall have seven stainless steel rods 3 inches long by 1/4-inch diameter (minimum) or 1/2-inch diameter aluminum rods equally spaced and welded, protruding upward at

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approximately a 45 degree angle. When tested as specified in 4.6.1, each large attachment peg shall be capable of supporting a 7-pound weight at its tip, and each small peg shall be capable of supporting a 3-pound weight at its tip.

* 3.5.4 Casters. Casters shall conform to type II, class A or B, wheel diameter 5 inches, style and mounting optional of FF-C-77. A 5-inch minimum diameter rotating rubber protective bumper shall be attached to each caster. Two of the four casters shall be provided with locking brakes.

3.6 Assembly instructions. Included with each rack shall be assembly instructions. These instructions shall, with the aid of illustrations, clearly describe how to assemble the rack.

3.7 Identification marking. Each rack shall be permanently and legibly marked with the manufacturer's name or trademark readily identifiable with said manufacturer.

* 3.7.1 Data name plates. The racks shall be furnished with a data name plate in accordance with the applicable requirements of MIL-STD-130 except the requirements for (a) Methods of applying, (b) Identification tags, (c) Information not required, and (d) Optional marking information shall not apply. The data name plate shall be made of minimum 20 gage corrosion resisting metal and attached to the rack by rivets, screws, or by welding, in such a manner as to meet the applicable National Sanitation Foundation sanitary requirements for this equipment. The plate shall contain the following information, which shall be stamped, engraved, or applied by photosensitive means:

National Stock Number
Procurement Instrument Identification Number
Specification Data
Manufacturer's Name, Address, and Telephone Number
Supplier's Name, Address, and Telephone Number (List only if different from Manufacturer)
Contract Number
Manufacturer's Model Number
Government Approved Manual Number (see 6.2)

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the rack.

3.8 Workmanship. The racks shall contain no sharp edges or corners, dents, scratches, burrs, or slivers and shall be free from dirt, smudge, grease, or other foreign matter. All bolt or rivet holes shall have burrs removed.

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Rivets shall be driven to completely fill the holes, with full size heads neatly finished and in full contact with the surface member. Welds shall be continuous and smooth and shall be free from porosity, cracks, burn holes, fractures, and excess flux or splatter. Lockwashers shall be used on all bolts, and threaded fasteners shall be drawn tight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3 and tested as specified in 4.6.1, 4.6.2, and 4.6.3. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.2 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of completely fabricated storage racks. The sample unit shall be one completely fabricated and assembled rack. The inspection level shall be S-4 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 10.0 for total combined major and minor defects.

TABLE I. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Finish	Scratches, dents	101	
	Dirty		201
Construction and workmanship (general)	Part or component missing or not specified type	102	
	Burrs, slivers, sharp edges, sharp corners	103	
Shelves	Not number specified	104	
Bumpers	Not type specified, missing	105	
Rivets	Not in full contact with surface of member		202
Casters	Not as specified	106	

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TABLE I. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Welds	Porous, fractured, burn holes, cracked, incomplete fusion	107	
	Not ground smooth to blend with adjacent surfaces, excess flux or splatter		203
Threaded fasteners	Incapable of being drawn up tight		204
	Lockwashers not provided for use with bolts		205
Identification marking	Missing, incomplete, illegible, not permanent	108	
Assembly instructions	Missing, incomplete, no illustrations, not clear		206
Data name plates	Omitted or not as specified		207
	Information incomplete or not legible		208
	Not located so as to be readily visible to operator		209

4.4.3 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified in 3.5 through 3.5.4. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of completely assembled storage racks. The sample unit shall be one rack. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4.4 End item testing. The end item shall be tested as specified in 4.6.1, 4.6.2, and 4.6.3. The lot size shall be expressed in completely assembled storage racks. The sample unit shall be one rack completely assembled in accordance with the manufacturer's instructions. Failure of a test shall be recorded as a defect. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container

* 4.5 Certification compliance examination. Certifications, certified test reports, or listing marks for codes and standards, as applicable, submitted in accordance with 3.3 shall be examined and validated as proof of compliance. Prior to the approval of the first article, if one is submitted, or prior to commencing production, the contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the racks he proposes to furnish under this document meet the applicable requirements of the National Sanitation Foundation (NSF).

4.5.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be one of the following:

a. A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished rack.

b. A certification for the rack issued by NSF under their special one-time contract evaluation/certification service.

c. A certified test report acceptable to the contracting officer with the advice of the Army Surgeon General from an independent testing laboratory, indicating that the racks have been tested and conform to the specified NSF standard.

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4.6 Methods of inspection.

4.6.1 Static load test. Hang a 3-pound load on the tip of each small attachment peg, a 7-pound load on the tip of each large attachment peg, and place a load of 120 pounds on each shelf and let stand for 10 minutes. Check for any breakage, permanent deformation, or parts separation for conformance with 3.5. Nonconformance with 3.5 shall constitute failure of this test.

4.6.2 Shock test. Distribute a load of 75 pounds on each shelf and push the loaded rack completely off a platform with a 1-inch vertical drop at a speed not less than 2 mph with the rack traveling perpendicularly to the step, seven times each end, for a total of fourteen trials. Check for parts breakage, separation, or permanent deformation, for conformance with 3.5. Nonconformance with 3.5 shall constitute failure of this test.

* 4.6.3 Racking test. Uniformly distribute a load of 75 pounds on the bottom shelf and another load of 75 pounds on the top shelf. With the caster wheels resting against a stop(s) to prevent rolling, momentarily apply a side force of 50 pounds perpendicular to the 24-inch dimension of the top shelf and parallel to the floor. Deflection at the point of force application shall not exceed the racking requirement of 3.5. Nonconformance with 3.5 shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. The storage rack shall be completely disassembled and attaching hardware shall be placed in a cloth drawstring bag. The four casters and bag containing attaching hardware shall be packed in a snug-fitting intermediate fiberboard box conforming to style RSC, type CF, variety DW, or type SF, class domestic of PPP-B-636. Box closure shall be in accordance with PPP-B-636.

* 5.1.2 Commercial. Each storage rack shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Each storage rack, preserved as specified in 5.1, shall be packed in the most compact manner in a cleated plywood shipping container conforming to overseas type, style A or B of PPP-B-601. The

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contents shall be immobilized as necessary to prevent movement or damage during transit. Each shipping container shall be closed and strapped in accordance with PPP-B-601.

5.2.2 Level B packing. Each storage rack, preserved as specified in 5.1, shall be packed in the most compact manner in a wood cleated fiberboard or cleated plywood shipping container conforming to class I, style A or B of PPP-B-591 or domestic type, style A or B, of PPP-B-601, respectively. The contents shall be immobilized as necessary to prevent movement or damage during transit. Each shipping container shall be closed in accordance with the applicable container document.

* 5.2.3 Commercial packing. Storage racks, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Marking. In addition to any special marking required by the contract or purchase order, interior packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The rack is intended to be used to store mixing bowls and attachments for the mixer specified in 00-M-38 Mixing Machines, Food, Electric (Vertical, Commercial Type, Bakery and Kitchen).

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Government approved manual number to be included on data name plate (see 3.7.1).
d. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.4).

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* 6.5 Subject term (key word) listing.

Kitchen equipment storage
Rack, mixer bowls

6.6 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

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Air Force - 99

Review activities:

Army - MD
Air Force - 84
DLA - GS

Preparing activity:

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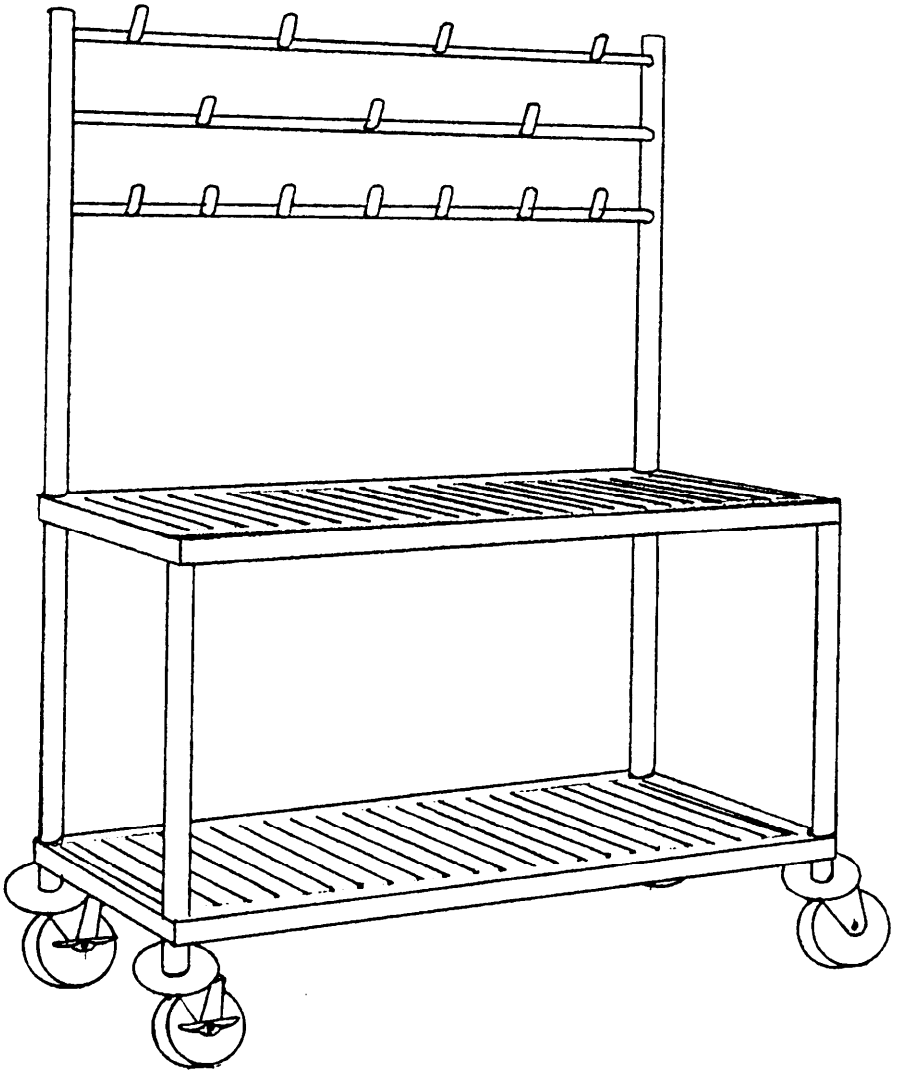


Figure 1. Rack, Storage, Mixer Bowls and Attachments

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-43870B	Rack, Storage, Mixer Bowls and Attachments, Mobile

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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